

PAPER_930: Production Scaling v7 Benchmark (300k calc/s)

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Abstract

Production benchmark suite targeting 300,000 calculations/second, a 3x improvement over the v4 baseline (100k calc/s). Implements 8 benchmark kernels: 26-Layer Gravity summation, $F_{U,Bi,i}$ Assembly, Phonon a_{res} computation, Jet $M_{jet}(\Gamma)$ evaluation, NS Spindown correction, GW190425 strain calculation, Full Pipeline v7 (all-in-one), and Vectorized Phonon Batch processing. Each kernel is timed independently and the aggregate throughput is compared against the 300k target. The benchmark validates that the new phonon physics modules (Session 211) maintain production-grade performance.

1. Core Equations

Section A: Benchmark Kernels

K1: $g = \sum_{k=1}^{26} \exp(-[SSq]*k/26) * (1 + 0.001*k)$
K2: $F_{U,Bi,i} = F_{\{U,Bi\}} * S_{26} * [SSq] * E_{net}$
K3: $a_{res} = (F_{\{U,Bi\}}/F_U) * \Phi * S_{26}$
K4: $M_{jet} = 1 + A_{jet} * \exp[-(\Gamma - \Gamma_0)^2 / (2 * \sigma^2)]$
K5: $\Omega_{dot_phonon} = \Omega_{dot} * (1 + \Phi * S_{26} * [SSq] / N)$
K6: $h_{UQFF} = h_{GR} * D_{total} * \exp([SSq]*t/26)$
K7: Full pipeline (K1-K6 sequential)
K8: Vectorized phonon batch (1000 frequencies)

Section B: Performance Metrics

Rate = $N_{iterations} / elapsed_time$ (calc/s)
Target: 300,000 calc/s aggregate
v4 baseline: 100,000 calc/s
Improvement factor: 3.0x

Section SM: SM Anchors

Python 3.14 runtime
Pure Python (no numpy requirement for core kernels)
Optional numpy vectorization for K8

2. Parameters

Parameter	Default	Description
n_iterations	10000	Benchmark iterations
target_calc_per_s	300000	Target throughput

3. Key Results

Kernel	Typical Rate (calc/s)	Status
26-Layer Gravity	~500k	PASS
Phonon a_res	~800k	PASS
Jet M_jet	~1M+	PASS
Full Pipeline	~200k	MARGINAL
Vectorized Batch	~400k	PASS

4. Physical Interpretation

The production scaling benchmark ensures that adding phonon physics modules does not degrade computational throughput below operational requirements. The 300k target represents the minimum rate needed for real-time simulation of 26-layer gravity across multiple astrophysical systems simultaneously (e.g., source2.cpp GUI parallel queries). Individual kernels exceed the target, but the full pipeline (all kernels sequential) approaches the limit, indicating that optimization of the pipeline serialization is the primary bottleneck.

5. References

- production_scaling_v4.py: Previous 100k baseline
- production_scaling_v7.py: Current 300k target implementation
- PAPER_922: M87 jet power curve (performance-critical MC sampling)